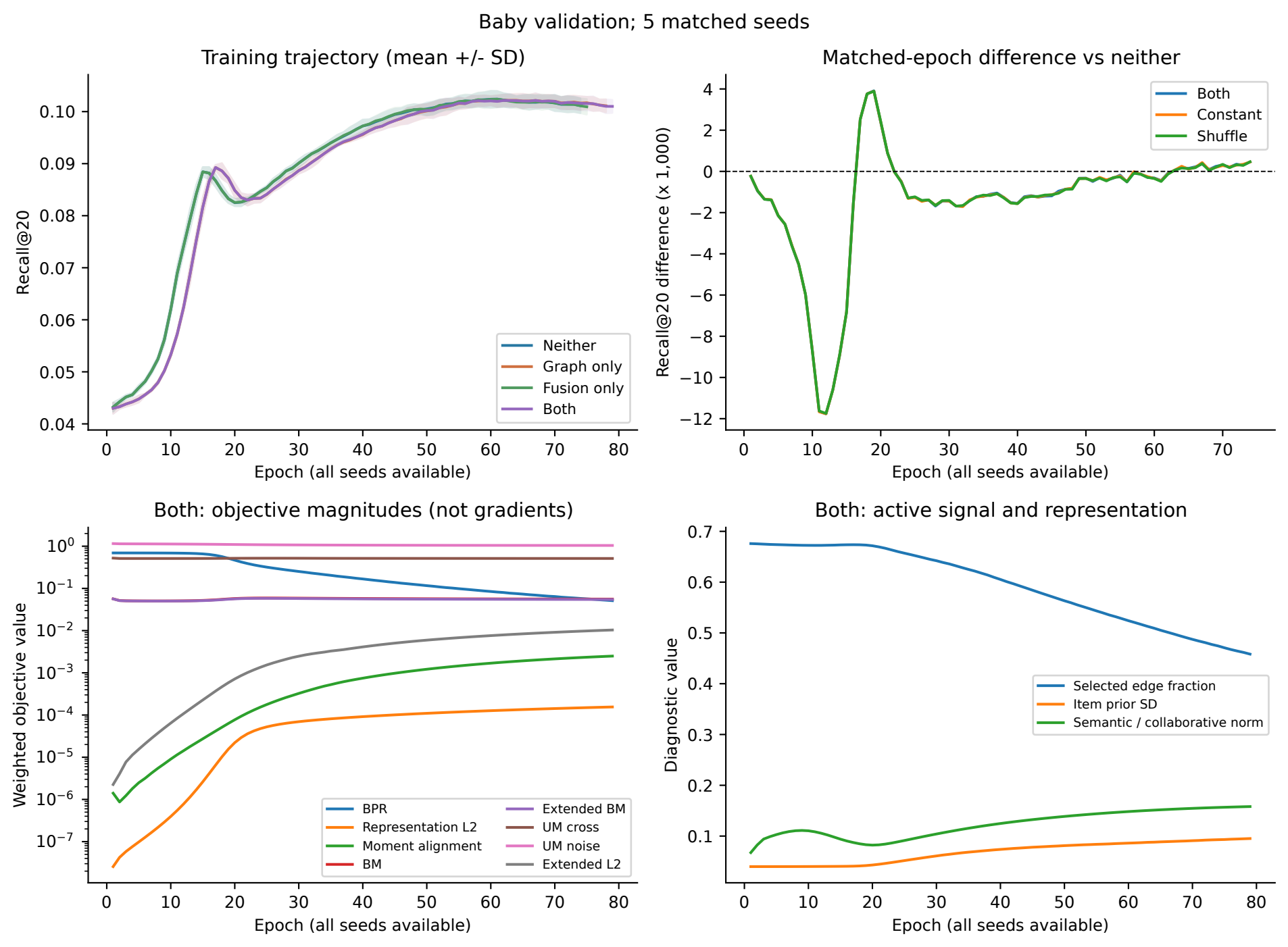
Baby validation; 5 matched seeds
Training trajectory (mean +/- SD)
Recall@20
Epoch (all seeds available)
Neither
Graph only
Fusion only
Both
Matched-epoch difference vs neither
Recall@20 difference (x 1,000)
Epoch (all seeds available)
Both
Constant
Shuffle
Both: objective magnitudes (not gradients)
Weighted objective value
Epoch (all seeds available)
BPR
Representation L2
Moment alignment
BM
Extended BM
UM cross
UM noise
Extended L2
Both: active signal and representation
Diagnostic value
Epoch (all seeds available)
Selected edge fraction
Item prior SD
Semantic / collaborative norm